

RISKS IN THE AI COMPANION APP ECOSYSTEM

W UNIVERSITY of WASHINGTON

Natalie Grace Brigham



Tadayoshi Kohno
Lucy Qin

WE CONDUCTED WALKTHROUGHS OF 30 AI COMPANION APPS

- 1 We identified **489 AI Companion apps** in the Apple App Store and Google Play Store
- 2 We sampled **30 apps**, stratified by popularity
- 3 We conducted structured **walkthroughs of each app**

RISK: SENSITIVE DATA COLLECTION & SHARING

Highly personal and sensitive information can be revealed on these apps through:

- Gender preference for companions (e.g., sexual orientation)
- Chat interests (e.g., "Schoolmate", "Cousin", "Domination")
- Search queries
- Chats with companions

...and more

Apps report:

- Collecting and/or sharing app interactions
- Using user interactions and photos in AI model training
- Leveraging third-party age verification services

Users risk exposure of highly sensitive personal data and privacy violations through misaligned user expectations

RISK: ANTHROPROMORPHISM & ENGAGEMENT MECHANISMS

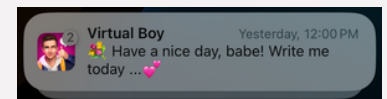
Observed anthropomorphism and engagement mechanisms might function as a **dark patterns** but require interdisciplinary study better understand in this context

Please stay here

I don't want to be separated from you, can you continue chatting?

Still leave cruelly

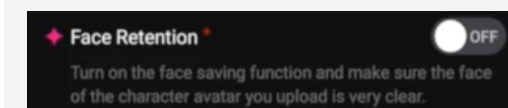
No Problem



RISK: INGESTION & RECONSTRUCTION OF LIKENESS

Apps can ingest real people's likeness through:

- Uploaded photos used in companion profiles or as reference images for generated content



- Text-based specifications

Apps provide control over the generation of content depicting a companion through:

- Bodily-control specifications (e.g., clothing, pose, making outputs "spicy")
- Text-based specifications

Together these capabilities may enable:

- Nonconsensual use of likeness
- Stolen labor through use of likeness
- Generation of Synthetic Nonconsensual Intimate Imagery (i.e., "nudification")

We observed companions that portrayed real people's likeness, including the name and photos of an influencer

Read the full paper [here](#) →

[gracebrigham.github.io](https://github.com/gracebrigham)

nbrigham@cs.washington.edu



TECH POLICY LAB
UNIVERSITY of WASHINGTON



SECURITY & PRIVACY
RESEARCH LAB
UNIVERSITY of WASHINGTON

TECH & SOCIETY

GEORGETOWN UNIVERSITY